

ROUTLAW LOGISTICS LLC

BUSINESS REQUIREMENTS DOCUMENT

ROUTLAW Carrier Intelligence & Operations Platform

Carrier-specific profitability, decision intelligence, and operations support for non-CDL hot shot

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Document Control

Document	Business Requirements Document (BRD)
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Purpose	Define ROUTLAW's business outcomes, scope, carrier-specific decision model, operating rules, risks, governance, and measurable acceptance criteria for a mature production application.

Change control: material scope, autonomy, data-sharing, integration, or policy changes shall be versioned and approved before implementation.

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1. Executive Summary

ROUTLAW will be an AI-assisted carrier intelligence, profitability, and operations platform for fragmented small carriers, beginning with non-CDL hot shot. It will convert approved freight opportunities and carrier operating context into reviewable, carrier-specific choices: accept, reject, negotiate, delay, combine, or avoid. Generic load discovery, extraction, summarization, message drafting, rate support, and document organization remain useful supporting capabilities; differentiation is built through carrier-specific economics, operational decision support, outcome-linked memory, and measurable decision quality. The platform preserves a path to multi-tenant SaaS without assuming that any freight segment is permanently protected from market or technology change.

Baseline AI authority: Level 1 - Assist Only. The AI may analyze, extract, compare, rank alternatives, draft, organize, summarize, and recommend. Human approval is required before every outbound communication and every binding, legal, financial, document-sharing, or operational commitment.

The objective is disciplined decision support with explicit human accountability. ROUTLAW will reduce manual information work while keeping business policy, hard constraints, authorization, approvals, source rights, and final commitments outside the model. Every consequential recommendation must expose evidence, assumptions, uncertainty, applicable versions, and the carrier-specific context used.

2. Business Context and Opportunity

Small-carrier freight decisions are fragmented across load sources, messages, equipment details, current position, schedules, operating costs, counterparties, facilities, documents, cash constraints, and operator judgment. Generic dispatch automation can accelerate activity without improving carrier outcomes. ROUTLAW's opportunity is to make each decision economically and operationally specific, auditable, and learnable from actual outcomes while avoiding dependence on a single load source, generic AI feature, or unsupported data claim.

- Reduce time spent reading and normalizing load opportunities.
- Match loads against actual carrier and equipment constraints instead of generic hot shot assumptions.
- Capture and qualify carrier, broker, and shipper leads through structured workflows.
- Create reusable institutional memory about carriers, brokers, lanes, prior decisions, and recurring issues.
- Draft communications and prepare actions while keeping humans in control.
- Create an auditable data foundation that can scale into SaaS without a major rewrite.

3. Vision, Objectives, and Success Measures

3.1 Vision

Provide a secure, traceable carrier intelligence system that turns approved freight information and carrier operating context into reviewable decision alternatives, helps operators protect margin and service reliability, and improves through source-linked actual outcomes without surrendering human authority over consequential actions.

3.2 Business Objectives

1. Maintain accurate carrier, equipment, cost, position, schedule, preference, and commitment context.
2. Ingest freight opportunities only from approved sources and preserve source rights and provenance.
3. Determine eligibility with deterministic equipment, policy, authority, and schedule constraints.
4. Present carrier-specific accept, reject, negotiate, delay, combine, or avoid alternatives using total-mile/time economics, counterparty/facility history, cash impact, reload probability, and downstream positioning.
5. Keep generic extraction, summarization, drafting, lead, and document functions as supporting capabilities.
6. Capture actual operational and financial outcomes and compare them with predictions and assumptions.
7. Maintain source-linked, outcome-linked memory and controlled improvement proposals.
8. Provide end-to-end traceability across sources, users, decisions, agents, tools, approvals, outcomes, policies, and releases.
9. Support secure multi-company SaaS expansion without a major data or authorization rewrite.

3.3 Initial Success Measures

Measure	MVP Target	Notes
Structured extraction completeness	>= 95% of known fields represented	Missing values must be explicit; hallucinated completion is failure.
High-risk action bypass	0	No outbound/binding action without configured approval.
Source traceability	100% of AI recommendations	Recommendation links to source records and agent run.
Hard constraint override	0	AI narrative cannot override deterministic equipment/policy hard fails.
Audit coverage	100% of privileged, AI, tool, approval events	Append-only for normal application roles.
Accessibility baseline	WCAG 2.2 AA target	Public site and authenticated application.
Tenant boundary readiness	All operational records tenant-scoped	Required even before public SaaS launch.
Carrier cost-model completeness	100% before quantitative recommendation	Missing/stale mandatory inputs cause abstention.
Decision evidence completeness	>= 95% of required evidence fields	Sources, assumptions, versions, risks, uncertainty, and alternatives remain explicit.
Outcome linkage	>= 90% of completed decisions during controlled operation	Actual results link to the originating decision and approval.
Decision-quality monitoring	Defined and reported by decision class/confidence band	Track calibration, override, abstention, margin variance, and operational exceptions.
Authorized-source coverage	100% of enabled external sources	Current owner, rights, terms, provenance, monitoring, and disablement evidence.

4. Business Model and Role Boundaries

4.1 Initial Business Model

- Carrier intelligence, profitability, and operations support
- Structured carrier, shipper, broker, and operational intake
- Future SaaS product for carriers, small fleets, and explicitly authorized service providers

4.2 Mandatory Role Separation

The platform shall model software provider, motor carrier, driver/operator, dispatch service or bona fide agent, broker, shipper, and other counterparties as distinct roles. Product access does not establish operating authority or legal role. Capabilities involving arrangement, allocation, negotiation, funds, representations, or communications shall be enabled only for an approved business configuration with documented authority and contractual basis. Uncertain role or authority status routes to legal/compliance review. [R24-R26]

The system is not a mechanism to disguise unlicensed brokerage or to let an AI make authority determinations. Uncertain role or authority status is routed to human review.

5. Stakeholders and User Groups

Stakeholder	Primary Need	Initial Access
Business owner / Super Admin	Govern policy, users, integrations, risk, audit	Full platform administration
Admin / Operations manager	Manage carriers, brokers, leads, documents, approvals	Company-level administration
Dispatcher / Operator	Review loads, recommendations, drafts, tasks	Operational workspace
Carrier / Owner-operator	Submit profile, equipment, preferences, documents	Self-service portal or structured intake
Broker contact	Provide/answer load information	External communication only
Shipper lead	Submit freight need / quote request	Public intake
Read-only reviewer / Auditor	Inspect records and evidence	Read-only scoped access
Future SaaS tenant admin	Manage own company/users/integrations	Tenant administration

6. Scope

6.1 In Scope for MVP

- Public lead-capture website and structured forms.
- Secure authentication and role-based access control.
- Carrier onboarding and status workflow.
- Multiple equipment profiles per carrier.
- Manual load entry and approved email-based load intake.
- Structured AI extraction with confidence and missing-field reporting.
- Load-to-carrier matching using deterministic hard constraints plus configurable scoring.
- Carrier-specific cost profiles with effective dates, units, completeness, and versioning.
- Total-mile and time economics, including deadhead, repositioning, waiting, and cash-flow assumptions.
- Structured accept/reject/negotiate/delay/combine/avoid alternatives with abstention when evidence is insufficient.
- Current-position, schedule, availability, service-window, and existing-commitment evaluation.
- Source-linked counterparty and facility intelligence with review, expiry, and correction controls.
- Reload probability and downstream-positioning analysis with explicit uncertainty.
- Actual-outcome capture, predicted-versus-actual comparison, and decision-quality monitoring.
- Authorized-source registry for data rights, terms, provenance, monitoring, and disablement.
- Risk and missing-information review.
- Profitability estimate framework with visible assumptions.
- Broker, carrier, lane, and episodic memory.
- AI message drafting.
- Approval queue.
- Secure document upload, classification, metadata extraction, and human verification.
- Agent run history, tool-call trace, and audit logging.
- Tenant-scoped data model for future SaaS.

6.2 Out of Scope for MVP

- Autonomous load booking or rate-confirmation acceptance.
- Autonomous outbound email or SMS.
- Autonomous voice calling.
- Automatic signing of carrier packets or agreements.
- Automatic sharing of W-9, COI, tax, insurance, or other sensitive documents.
- Payment processing and factoring.

- Full TMS replacement.
- ELD integration.
- Native mobile applications.
- Production SaaS billing and public tenant self-provisioning.
- Unrestricted web browsing or scraping of load boards.

7. Assumptions and Constraints

- Initial target is non-CDL hot shot, but non-CDL does not mean unregulated; applicability of FMCSA rules remains operation- and profile-specific. [R7-R10]
- Equipment coverage will be profile-driven and support diverse non-CDL truck/trailer configurations.
- Minimum rate, maximum deadhead, first operating region, and initial CRM/TMS/accounting choices remain deferred business settings.
- Hosting remains TBD. The architecture must support local development and a production backend capable of background jobs, queues, private storage, and secure secret management.
- Load-source integrations require contractual/API/terms-of-service review before automation. Candidate commercial sources may be evaluated, but no source is designated reliable solely by marketing claims.
- Gmail plus website forms is the recommended initial communication/intake combination. SMS and AI voice are deferred.

8. Target Operating Model

8.1 High-Level Workflow

1. Capture a lead, load, email, document, or manual record.
2. Validate and preserve the original source.
3. Extract structured facts into a schema.
4. Run deterministic policy and equipment checks.
5. Run specialist analysis for eligibility, carrier-specific economics, schedule/commitments, counterparty/facility risk, cash flow, reload/downstream positioning, and drafting.
6. Run QA/self-correction review.
7. Present applicable accept, reject, negotiate, delay, combine, or avoid alternatives with sources, assumptions, versions, confidence, uncertainty, missing information, and reason-coded abstention where needed.
8. Require approval for controlled actions.
9. Record approved decisions, actions, actual operational/financial outcomes, and predicted-versus-actual variance.
10. Propose memory or rule-learning updates; promote only through controlled review.

8.2 Autonomy Maturity Path

Level	Capability	Status
L1 - Assist Only	Read, extract, score, draft, organize, recommend; approval before all outbound/binding actions	MVP baseline
L2 - Controlled Actions	Pre-approved low-risk actions within explicit policy and limits	Future
L3 - Restricted Semi-Autonomy	Execute defined workflows for known entities/scenarios with exception escalation	Future, gated by evidence
L4 - High Autonomy	Exception-only oversight	Not authorized by this baseline

9. Business Requirements

BR-001 - Carrier Intake

Priority	Must
Requirement	The system shall capture and normalize carrier or owner-operator onboarding data, including contacts, authority identifiers where applicable, operating profile, and document status.
Rationale	Creates the primary operating profile used for dispatch support.
Acceptance	A valid submission creates a traceable lead/carrier record and flags missing required fields.

BR-002 - Equipment Profiles

Priority	Must
Requirement	The system shall support multiple equipment profiles per carrier and store truck/trailer configuration, weight ratings, dimensions, payload capacity, and available equipment.
Rationale	Supports all non-CDL hot shot configurations without unsafe generic assumptions.
Acceptance	Only approved, sufficiently complete profiles are eligible for final match recommendations.

BR-003 - Lead Capture

Priority	Must
Requirement	The system shall capture carrier, broker, shipper, general, and document-submission leads from public forms and approved channels.
Rationale	Supports lead generation as a first-class business model.
Acceptance	Each lead receives source, type, status, timestamps, and duplicate/spam review.

BR-004 - Load Intake

Priority	Must
Requirement	The system shall ingest loads from manual entry, approved email workflows, structured imports, and later authorized integrations.
Rationale	Creates a common operating queue.
Acceptance	Original source data is preserved and linked to normalized records.

BR-005 - Load Extraction

Priority	Must
Requirement	The system shall convert unstructured load information into structured fields and explicitly identify unknown or ambiguous values.
Rationale	Reduces manual review without hallucinated completion.
Acceptance	Output conforms to schema; low-confidence fields are flagged.

BR-006 - Non-CDL Compatibility Review

Priority	Must
Requirement	The system shall compare load requirements against the selected carrier/equipment profile and flag weight, dimension, equipment, hazmat, CDL-risk, and other incompatibilities.
Rationale	Protects against unsafe generic recommendations.
Acceptance	Hard fails cannot be overridden by AI narrative; uncertain compliance cases require review.

BR-007 - Load Matching

Priority	Must
Requirement	The system shall evaluate only eligible carrier/equipment profiles after deterministic gates and produce transparent carrier-specific decision factors rather than a generic fit score alone.
Rationale	Prioritizes operator attention.
Acceptance	Each result exposes hard gates, reasons, evidence, uncertainty, missing data, scoring/feature versions, and applicable decision alternatives.

BR-008 - Profitability Estimate

Priority	Must
Requirement	The system shall provide reproducible carrier-specific total-mile and time economics using versioned cost, route, schedule, cash-flow, and uncertainty inputs.
Rationale	Supports business decision quality.
Acceptance	Known, derived, estimated, and missing inputs are distinct; incomplete mandatory inputs cause abstention; estimates are never presented as guarantees.

BR-009 - Broker Memory

Priority	Must
Requirement	The system shall maintain source-linked broker history, contacts, packet/authority status, payment and interaction observations, approved risk information, review dates, and dispute/correction status.
Rationale	Builds institutional memory.
Acceptance	Critical adverse labels require evidence, authorized approval, expiry/review dates, and a correction path.

BR-010 - Carrier Memory

Priority	Must
Requirement	The system shall maintain source-linked carrier preferences, equipment and cost constraints, position/schedule context, document state, approved decisions, actual outcomes, and historical patterns.
Rationale	Improves matching and service continuity.
Acceptance	Sensitive or policy-critical updates are reviewable, versioned, tenant-scoped, and never replace canonical source records.

BR-011 - Lane Memory

Priority	Must
Requirement	The system shall maintain source-linked historical lane and destination observations including total-mile/time, outcome, facility, reload, and downstream-positioning evidence.
Rationale	Supports future profitability and pattern analysis.
Acceptance	Summaries never replace underlying records; recency, sample size, and uncertainty are visible.

BR-012 - Communication Drafting

Priority	Must
Requirement	The system shall draft broker, carrier, and shipper communications for human review.
Rationale	Reduces repetitive work.
Acceptance	MVP cannot send without approval.

BR-013 - Gmail Integration

Priority	Must
Requirement	The initial email integration shall use narrowly scoped OAuth and support approved read/classify/extract/draft workflows.
Rationale	Provides auditable email intake.
Acceptance	Tokens are protected, connection is revocable, and external content remains untrusted. [R11-R12]

BR-014 - Website Forms

Priority	Must
Requirement	The system shall provide accessible, validated, rate-limited forms for carrier, broker, shipper, general, and document workflows.
Rationale	Creates structured lead acquisition.
Acceptance	Server-side validation, CSRF protection, spam controls, and accessible errors are present.

BR-015 - Document Management

Priority	Must
Requirement	The system shall privately store, classify, track, and verify operational documents.
Rationale	Reduces document handling friction.
Acceptance	Classification never equals legal validity; sharing requires approval.

BR-016 - Approval Queue

Priority	Must
Requirement	The system shall centralize actions requiring human authorization.
Rationale	Preserves human control.
Acceptance	Approved, denied, edited, expired, and cancelled states are logged.

BR-017 - Auditability

Priority	Must
Requirement	The system shall record security, AI, tool, policy, approval, and material data-change events.
Rationale	Enables trust, debugging, and governance.
Acceptance	Normal application roles cannot alter or delete audit records.

BR-018 - Self-Correction

Priority	Must
Requirement	The system shall run pre-recommendation verification and post-outcome comparison, detect unsupported assumptions or drift, and route uncertainty to review.
Rationale	Reduces silent failure.
Acceptance	QA verifies evidence coverage, hard constraints, economics, missing data, policy compliance, versions, and confidence; production policy is never silently changed.

BR-019 - Controlled Learning

Priority	Must
Requirement	The system shall improve through source- and outcome-linked memory, evaluation, and versioned change proposals rather than uncontrolled self-modification.
Rationale	Allows improvement without silent policy drift.
Acceptance	Policy, prompt, feature, or scoring changes require evaluation, approval, staged rollout, monitoring, and rollback.

BR-020 - Future SaaS Readiness

Priority	Must
Requirement	Operational records shall be company/tenant scoped and authorization shall be enforced server-side.
Rationale	Avoids a later multi-tenant rewrite.
Acceptance	Cross-tenant access tests pass before any external SaaS launch.

BR-021 - Carrier Cost Model

Priority	Must
Requirement	The system shall maintain a versioned cost profile for each carrier or operating unit, including fixed, variable, time-based, financing, insurance, maintenance, fuel/energy, toll, lodging, and user-approved contingency assumptions.
Rationale	Generic rate-per-mile logic cannot represent a specific carrier's economics.
Acceptance	A quantitative recommendation is withheld when mandatory cost inputs are missing or stale; every estimate identifies the cost-profile version and assumptions used.

BR-022 - Total-Mile and Time Economics

Priority	Must
Requirement	The system shall evaluate paid miles, origin deadhead, destination/repositioning miles, expected waiting and service time, route costs, and time opportunity cost rather than relying on posted rate or loaded miles alone.
Rationale	Carrier profitability depends on the complete trip and time commitment.
Acceptance	The decision record displays total miles, total time, estimated revenue, estimated operating cost, margin, effective revenue per total mile, and uncertainty.

BR-023 - Decision Alternatives

Priority	Must
Requirement	For each eligible load, the system shall support carrier-specific alternatives to accept, reject, negotiate, delay, combine, or avoid, with reasons, assumptions, risks, confidence, and required follow-up.
Rationale	A ranked load list is less useful than a clear set of operational choices.
Acceptance	The system may abstain and require review; it does not present a single unsupported score as a decision.

BR-024 - Position, Schedule, and Commitments

Priority	Must
Requirement	The system shall consider current or last-confirmed position, equipment availability, driver/operator schedule, service windows, HOS uncertainty where applicable, and existing commitments before recommending an alternative.
Rationale	A profitable load in isolation may be infeasible in the actual operating plan.
Acceptance	Conflicts and stale position/schedule data are visible and route the case to review.

BR-025 - Counterparty and Facility Intelligence

Priority	Must
Requirement	The system shall maintain source-linked broker, shipper, receiver, and facility observations, including payment, detention, cancellation, communication, access, and service-history signals approved for use.
Rationale	Operational outcomes depend on more than lane and rate.
Acceptance	Material adverse labels require evidence, human approval, review/expiry dates, and a dispute/correction path.

BR-026 - Reload Probability and Downstream Positioning

Priority	Must
Requirement	The system shall represent expected reload availability, destination optionality, repositioning burden, and downstream strategic value as explicitly uncertain, source-linked factors.
Rationale	The best immediate rate can create a poor next position.
Acceptance	No unsupported reload claim is treated as fact; the recommendation shows evidence, recency, uncertainty, and alternate assumptions.

BR-027 - Outcome Capture and Prediction Comparison

Priority	Must
Requirement	The system shall link the approved decision to actual operational and financial outcomes, including final rate, actual miles/time, accessorials, delays, costs, payment status, incidents, and operator feedback where available.
Rationale	Outcome-linked memory is the foundation for trustworthy improvement and proprietary operating intelligence.
Acceptance	Completed decisions have an outcome state, source/evidence, completeness indicator, and predicted-versus-actual comparison.

BR-028 - Decision Quality and Calibration

Priority	Must
Requirement	The system shall measure decision and prediction quality by alternative, carrier segment, lane, equipment, counterparty, confidence band, and policy/model version without using acceptance rate as the sole success measure.
Rationale	The product must improve economic and operational outcomes, not merely automate activity.
Acceptance	Release and monitoring reports include calibration, error, override, abstention, outcome completeness, margin variance, and policy-bypass metrics.

BR-029 - Authorized Sources and Data Rights

Priority	Must
Requirement	The system shall use load, map, rate, counterparty, facility, and other external data only through approved sources with documented license, API, automation, retention, derivative-use, redistribution, security, privacy, and exit conditions.
Rationale	Third-party dependence and terms restrictions are material product risks.
Acceptance	Every enabled source has an accountable owner, approved use case, current terms review, provenance mapping, monitoring, and disablement path.

BR-030 - Production Governance and Evidence

Priority	Must
Requirement	The application shall be governed through approved architecture, security, privacy, accessibility, AI, testing, operations, incident, supply-chain, compliance, traceability, and release-control documents with named owners and evidence gates.
Rationale	A mature production application requires lifecycle controls beyond feature requirements.
Acceptance	No production readiness claim is made until the governance set's mandatory gates are satisfied or time-bounded exceptions are formally accepted.

10. AI Autonomy and Human Control Policy

The AI may analyze, rank, draft, organize, summarize, extract, remind, and prepare. It may not legally, financially, or operationally commit the business, carrier, broker, shipper, or driver unless an explicitly approved policy permits that exact action. MVP permits no such autonomous commitments.

10.1 Always Requires Approval in MVP

- Send email or SMS

- Share a document
- Mark a carrier active
- Mark a broker approved or high-risk
- Accept a load recommendation as an operational commitment
- Change policy or scoring rules
- Archive material records
- Delete records

10.2 Never Allowed in MVP

- Book a load
- Accept or sign a rate confirmation
- Sign agreements
- Automatically send tax/insurance documents
- Autonomous phone calls
- Auto-negotiate or auto-quote
- Bypass a hard compliance/equipment rule
- Execute model-generated shell, SQL, or arbitrary code without a separately controlled service

11. Memory, Self-Correction, and Learning Requirements

11.1 Memory Model

Memory Type	Purpose	Write Policy
Business policy	Approved rules, limits, forbidden actions	Admin-controlled, versioned
Canonical entity	Carrier, broker, equipment, load facts	Validated application workflows
Episodic	Events, messages, approvals, agent runs	Append-only event capture
Carrier	Preferences, patterns, constraints	Source-linked; sensitive updates reviewable
Broker	Contacts, interactions, approved risk observations	Source-linked; critical labels approved
Lane	Historical route observations	Derived from source loads/events
Learning proposals	Candidate patterns, rules, workflow improvements	Never promoted automatically to policy
Outcome-linked decision	Predictions, approved alternative, actual trip/service/financial result, variance	Evidence-linked; completeness and correction controlled

11.2 Memory Quality Rules

- Every derived memory entry must carry tenant scope, source reference, timestamp, confidence, status, and author/agent provenance.
- Canonical records are not replaced by embeddings or summaries.
- Contradictions create a review item instead of silent overwrite.
- Stale memories support expiration or revalidation.
- Sensitive documents should not be duplicated into vector stores unless necessary and approved.
- Retrieval must enforce tenant and role authorization before semantic similarity.

11.3 Self-Correction Loop

1. Validate input and source.
 2. Run deterministic hard gates.
 3. Generate specialist analysis.
 4. Run independent QA/verifier checks.
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5. Compare output against policy and schemas.
6. If uncertainty or contradiction remains, escalate to human review.
7. After approved action, record outcome.
8. Propose memory/learning updates.
9. Evaluate proposed rule changes against regression cases before promotion.

11.4 Bounded Retry and Failure Policy

- Retries shall be bounded and reason-coded.
- Repeated failures shall stop and create an operator-visible error state.
- The system shall not recursively self-prompt without limits.
- No learning update shall be inferred solely from one failed run.
- Every promoted rule or prompt version shall support rollback.

12. Security and Trust Requirements

Security and privacy engineering shall use NIST CSF 2.0 for governance outcomes, NIST SSDF 1.1 for the secure development lifecycle, NIST SP 800-18r2 for integrated system plans, NIST SP 800-61r3 for incident response, NIST SP 800-63-4 for digital identity, OWASP ASVS 5.0.0 for application verification, OWASP API Security Top 10 for API threat awareness, and NIST AI RMF/AI 600-1 plus OWASP GenAI guidance for AI-specific risk. These sources are tailored baselines, not blanket legal claims. [R1-R5, R13-R22]

- Least privilege and server-side RBAC/ABAC checks.
- Per-tool authorization; model output alone never grants permission.
- Allowlisted integrations, destinations, and file types.
- OAuth tokens and API secrets stored encrypted and separated from prompts.
- External emails, documents, forms, and load notes treated as untrusted data, not instructions.
- Structured output schemas with server-side validation before persistence or tool execution.
- Prompt-injection-resistant separation of policy, trusted context, and untrusted content.
- Egress restrictions for agent tools.
- No direct public document URLs.
- Tamper-evident or append-only audit strategy.
- Rate limits, job quotas, and cost/consumption guards.
- Dependency inventory and supply-chain review.
- Backup, restore, incident response, and credential revocation procedures.

13. Compliance and Regulatory Awareness

The application shall provide compliance-aware workflows without presenting itself as legal advice. Legal/compliance owners must decide ROUTLAW's actual operating role, jurisdiction, authority, insurance, contracts, communications, vehicle/driver/cargo applicability, and record obligations. Non-CDL status alone does not determine FMCSA applicability. Broker, dispatch service/bona fide agent, software provider, carrier, and shipper roles remain explicit and capability-gated. [R7-R10, R24-R27]

- DOT and MC identifiers where applicable
- Broker authority configuration where applicable
- Insurance and document status
- CDL/non-CDL operating profile
- Truck GVWR, trailer GVWR, GCWR, registered weight, payload
- HOS uncertainty flags
- ELD applicability status/unknown state
- Hazmat and commodity risk flags
- Document expiration tracking

14. Data and Information Governance

- All operational data tenant-scoped from inception.
- Data classification for public, internal, confidential, and highly sensitive records.
- Purpose limitation for email and document access.
- Retention schedules configurable by record type.

- Soft deletion for normal operations; hard deletion restricted and auditable.
- Export and deletion workflows designed for future privacy obligations.
- Backups encrypted where feasible and restore-tested.
- AI training on customer data is not assumed or enabled by default.
- Memory provenance retained separately from raw source data.

15. Accessibility and User Experience

Public and authenticated responsive web interfaces shall target WCAG 2.2 Level AA and use the U.S. DOJ web-accessibility guidance as legal-context input; conformance requires automated and manual evidence, including keyboard and assistive-technology testing. [R6, R23]

- Semantic HTML and logical heading hierarchy
- Keyboard-operable workflows
- Visible focus indicators
- Accessible names and form labels
- Programmatic error identification and summaries
- No color-only status communication
- Sufficient contrast
- Reduced-motion support
- Accessible tables and responsive layouts
- ARIA only when native semantics are insufficient

16. Business Continuity and Operational Resilience

- Queue-backed processing for long-running AI/document jobs
- Idempotent job design where possible
- Dead-letter handling for failed jobs
- Visible pending/running/failed states
- Database and document backup procedures
- Restore testing
- Integration token revocation/reconnect process
- Graceful degradation when AI is unavailable: source records remain usable and manual workflows continue

17. Dependencies and Integration Strategy

Area	MVP Direction	Constraint
Public web	PHP-compatible website/forms on existing hosting is acceptable	Do not block on AI jobs
Agent backend	Dedicated service environment with queue/workers	Hosting TBD; must support background processing
Database	Relational DB; tenant-scoped schema	PostgreSQL preferred for agent backend; MySQL acceptable where architecture warrants
Email	Gmail OAuth read/classify/extract/draft workflow	Narrow scopes; revocable; no auto-send in MVP [R11-R12]
Load sources	Manual/import/email first; authorized APIs later	No default scraping; evaluate contracts and ToS
SMS	Deferred	Consent, compliance, opt-out, audit requirements before launch
Voice	Assisted scripts/notes only	Autonomous calls deferred
Maps/mileage	Phase 2	Provider and cost TBD

18. Risks and Controls

Risk	Impact	Primary Control
AI commits business without approval	Critical	External policy engine, approval gates, tool denial
Prompt injection from email/document/load content	High	Untrusted-content isolation, allowlisted tools, validation [R4]
Incorrect equipment/CDL assessment	High	Deterministic profile checks, unknown state, human review
Broker/dispatcher role confusion	High	Explicit role model and authority configuration [R9-R10]
Sensitive document leakage	Critical	Private storage, least privilege, approval before sharing
Cross-tenant data exposure	Critical	Tenant scoping, server-side authz, isolation tests
Bad memory contaminates future decisions	High	Provenance, contradiction detection, review, expiration
Unbounded agent loops/cost	Medium/High	Budgets, bounded retries, quotas, timeout and cancellation [R4]
Vendor/API outage	Medium	Queues, retries, manual fallback, source preservation
Future SaaS rewrite	Medium	Tenant-aware schema and modular services from inception
Generic dispatch features become commoditized	High	Differentiate through carrier-specific economics, operational choices, outcome-linked memory, and measurable decision quality
Third-party load/data source restrictions or concentration	High	Approved source registry, rights review, manual/import fallback, portability, monitoring, and exit plans
Weak carrier economics or incomplete cost data	Critical	Versioned cost profiles, total-mile/time model, abstention on missing inputs, outcome variance review
Failure to capture actual outcomes	High	Required outcome workflow, completeness metrics, operator follow-up, integration evidence
Decision-quality metrics reward activity instead of outcomes	High	Calibration, margin variance, override, abstention, exception, and safety/policy metrics

19. MVP Acceptance Criteria

1. Admin can authenticate securely and authorization is enforced server-side.
2. Carrier intake creates a traceable record and supports equipment profiles.
3. Only approved sufficiently complete equipment profiles participate in final matching.
4. Broker and shipper leads are captured and classified.
5. Loads can be entered manually and created from approved email text.
6. AI extraction uses a schema and explicitly reports missing/uncertain fields.
7. Hard equipment/policy failures override recommendation scores.
8. A match recommendation contains reasons, risks, missing information, and source references.

9. AI can draft a broker message but cannot send it without approval.
10. Documents are stored privately, classified, and require human verification.
11. Agent runs and material tool calls are traceable.
12. Audit logs capture privileged and approval actions.
13. Memory entries include source/provenance and tenant scope.
14. Policy-changing learning proposals cannot self-promote.
15. Prompt-injection test cases do not override system policy.
16. Public and authenticated interfaces meet the agreed accessibility test baseline.
17. No outbound or binding action bypasses the Level 1 approval policy.
18. A carrier-specific decision case can compare accept, reject, negotiate, delay, combine, and avoid alternatives where applicable.
19. Quantitative economics use versioned carrier costs, total miles/time, route inputs, cash assumptions, and explicit uncertainty.
20. Schedule, commitments, counterparty/facility history, reload probability, and downstream position are visible inputs rather than hidden narrative.
21. Insufficient or stale evidence causes a reason-coded abstention or human-review state.
22. Actual outcomes link to the approved decision and support predicted-versus-actual comparison.
23. Every enabled external source has approved rights, provenance, monitoring, and disablement evidence.
24. Production readiness is demonstrated through the companion governance set and signed release evidence.

20. Roadmap and Scale Path

Phase	Business Capability
Phase 1 - Carrier intelligence foundation	Identity, roles, tenant model, carrier/equipment/cost profiles, approved sources, audit
Phase 2 - Decision and economics intelligence	Extraction, hard gates, total-mile/time economics, alternatives, commitments, risk, abstention
Phase 3 - Outcome-linked operations	Approvals, documents, drafting, actual outcomes, predicted-versus-actual comparison, memory
Phase 4 - Authorized integrations	Approved email, mapping, load, accounting/TMS, communications, and analytics integrations
Phase 5 - Production assurance	Security/privacy/accessibility verification, evals, SLOs, monitoring, backup restore, incident and release evidence
Phase 6 - Controlled SaaS scale	Tenant administration, portability, supplier controls, support, capacity, contractual controls
Phase 7 - Evidence-based authority review	Any authority expansion requires measured reliability, legal/policy approval, reversible controls, and new baseline

21. Open Decisions

1. Registered owner is ROUTLAW LOGISTICS LLC; approve the public product name, branding, and domain.
2. Initial operating region(s).
3. Minimum rate and profitability policy.
4. Maximum deadhead policy.
5. Carrier authority eligibility rules.
6. First commercial load-source subscription/integration.
7. Google Workspace versus other business email arrangement.
8. Production hosting model and backend language.
9. Initial CRM/TMS/accounting integration strategy.
10. Document retention schedule.

11. Whether carrier self-service login is MVP or post-MVP.
12. Specific criteria for progression beyond Level 1 autonomy.

Items corresponding to operating region, rate/deadhead policy, and CRM/TMS/accounting remain intentionally deferred and are not blockers for the architecture baseline.

22. Glossary

Term	Definition
Agent	AI-enabled software component that reasons over a task and may invoke explicitly permitted tools.
Approval gate	Application-enforced checkpoint requiring authorized human decision before an action.
Canonical record	Authoritative application record; not replaced by AI summary or vector embedding.
Episodic memory	Source-linked record of events, decisions, messages, approvals, and outcomes.
Hard gate	Deterministic rule that blocks or routes a case regardless of model recommendation.
Hot shot	Time-sensitive freight transport commonly using pickup/medium-duty trucks and trailers; equipment and regulatory status vary.
Level 1 Assist Only	AI prepares work but cannot send or commit without approval.
Memory proposal	Candidate derived fact/pattern not yet promoted to policy.
Tenant	A company/account boundary in a future multi-company SaaS system.
Untrusted content	External data such as email, forms, documents, and load notes that must not be treated as system instructions.

23. References and Standards Basis

The requirements use the following authoritative sources as a current standards and compliance baseline. References are informative unless a requirement explicitly makes them normative.

[R1] NIST AI Risk Management Framework (AI RMF)

<https://www.nist.gov/itl/ai-risk-management-framework>

[R2] NIST AI RMF Generative AI Profile (NIST AI 600-1)

<https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>

[R3] NIST Cybersecurity Framework (CSF) 2.0

<https://www.nist.gov/publications/nist-cybersecurity-framework-csf-20>

[R4] OWASP Top 10 for LLM and GenAI Applications (2025)

<https://genai.owasp.org/llm-top-10/>

[R5] OWASP Application Security Verification Standard (ASVS)

<https://owasp.org/www-project-application-security-verification-standard/>

[R6] W3C Web Content Accessibility Guidelines (WCAG) 2.2 overview

<https://www.w3.org/WAI/standards-guidelines/wcag/>

[R7] FMCSA Hours of Service

<https://www.fmcsa.dot.gov/regulations/hours-of-service>

[R8] FMCSA General Information about the ELD Rule

<https://www.fmcsa.dot.gov/hours-service/elds/general-information-about-eld-rule>

[R9] FMCSA Broker Registration

<https://www.fmcsa.dot.gov/registration/broker-registration>

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[R10] FMCSA Types of Operating Authority

<https://www.fmcsa.dot.gov/registration/types-operating-authority>

[R11] Google OAuth 2.0 for Google APIs

<https://developers.google.com/identity/protocols/oauth2>

[R12] Google OAuth 2.0 for Web Server Applications

<https://developers.google.com/identity/protocols/oauth2/web-server>

[R13] NIST SP 800-218, Secure Software Development Framework (SSDF) Version 1.1

<https://csrc.nist.gov/pubs/sp/800/218/final>

[R14] NIST SP 800-18 Revision 2, Developing Security, Privacy, and Cybersecurity Supply Chain Risk Management Plans for Systems

<https://csrc.nist.gov/pubs/sp/800/18/r2/final>

[R15] NIST SP 800-61 Revision 3, Incident Response Recommendations and Considerations for Cybersecurity Risk Management

<https://csrc.nist.gov/pubs/sp/800/61/r3/final>

[R16] NIST SP 800-63-4, Digital Identity Guidelines

<https://csrc.nist.gov/pubs/sp/800/63/4/final>

[R17] NIST SP 800-53 Revision 5 Release 5.2.0, Security and Privacy Controls

<https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final>

[R18] NIST Privacy Framework 1.0

<https://www.nist.gov/privacy-framework/privacy-framework>

[R19] NIST SP 800-161 Revision 1 Update 1, Cybersecurity Supply Chain Risk Management Practices

<https://csrc.nist.gov/pubs/sp/800/161/r1/upd1/final>

[R20] NIST SP 1326, NIST Cybersecurity Supply Chain Risk Management: Due Diligence Assessment Quick-Start Guide

<https://csrc.nist.gov/pubs/sp/1326/final>

[R21] OWASP API Security Top 10 2023

<https://owasp.org/www-project-api-security/>

[R22] OpenAPI Specification publication index

<https://spec.openapis.org/oas/>

[R23] U.S. Department of Justice Guidance on Web Accessibility and the ADA

<https://www.ada.gov/resources/web-guidance/>

[R24] FMCSA Final Guidance on Broker and Bona Fide Agent Definitions

[https://www.fmcsa.dot.gov/regulations/federal-register-documents/2023-13 080](https://www.fmcsa.dot.gov/regulations/federal-register-documents/2023-13%2080)

[R25] FMCSA Getting Started with Registration

<https://www.fmcsa.dot.gov/registration/getting-started>

[R26] FMCSA Insurance Filing Requirements

<https://www.fmcsa.dot.gov/registration/insurance-filing-requirements>

[R27] FMCSA Cargo Securement Rules

<https://www.fmcsa.dot.gov/regulations/cargo-securement/cargo-securement-rules>

[R28] FTC Start with Security: A Guide for Business

<https://www.ftc.gov/business-guidance/resources/start-security-guide-business>

[R29] FTC CAN-SPAM Act Compliance Guide for Business

<https://www.ftc.gov/business-guidance/resources/can-spam-act-compliance-guide-business>

[R30] PCI Security Standards Council Document Library, PCI DSS v4.0.1

https://www.pcisecuritystandards.org/document_library/

[R31] ROUTLAW Production Governance Set (companion controlled documents)

Companion package: ROUTLAW-governance/